

Jiaqi (Victoria) Duan

AI Systems Engineer — Agent systems, evaluation, retrieval, and developer tools

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Skills

Python, TypeScript/JavaScript, LLM Evaluation, Agent Orchestration, Retrieval, Embeddings, **FastAPI**, Django, Node.js, **PostgreSQL**, Redis, React, Next.js, Docker, Google Cloud Platform (GCP)

Experience

Founder & AI Systems Engineer, Siltpoke – Remote Apr 2026 – Present

- Built Siltpoke, an **independent reviewer for coding agents** using code changes, tool traces, and project context
- Grounded review findings in **verbatim tool output**, rejecting claims without supporting execution output
- Benchmarked **graph vs. text retrieval** on 9 planted bugs; text retrieval improved recall from **2/9** to **6/9**
- Wrote **13** regression eval suites for review quality, retrieval, and memory, plus an end-to-end pipeline evaluation

Founding Engineer, Ripplet – Remote Jun 2024 – Aug 2025

- Led engineering for a **HIPAA-aware, AI-assisted therapist platform prototype** with a cross-functional team
- Developed a **multi-agent RAG pipeline** linking session transcripts to evidence-based psychological frameworks
- Translated psychologist and stakeholder feedback into technical requirements and implementation decisions

Full-Stack Engineer & Dev Lead, Tech4Good Lab – Santa Cruz, CA Jun 2023 – Jan 2025

- Led engineers and designers building Pathways, an **AI learning platform** piloted with **100+ university students**
- Reduced staging page load times by **35%** in a Solid.js/Firebase app through SSR, lazy loading, and API caching
- Designed and **A/B-tested LLM prompt variants** to mitigate hallucinations, improving task completion by **15%**

Projects

Project Life Cycle | Python, JavaScript, Shell Apr 2026 – Present

- Built an **agent harness** orchestrating researcher, builder, verifier, and reviewer agents from initial spec to release
- Abstracted **host differences** to support one lifecycle across five coding hosts, including Claude Code and Codex
- Implemented a generator for **project-specific agent rules and guardrails** based on each codebase's detected stack

Travel Agent | Python, AutoGen, Ollama, Redis Apr 2025 – Nov 2025

- Developed a **multi-agent pipeline** with web/API retrieval, **critic-driven refinement**, and controlled fallback
- Integrated **Redis-backed session state** to share evidence, intermediate outputs, and feedback across agents
- Ran a **four-configuration ablation** across 10 user cases; critic scores correlated with human ratings ($r \approx 0.54$)

Yi | React, TypeScript, Next.js, Django, PostgreSQL Apr 2024 – Sep 2024

- Built and deployed a **B2B operations platform** supporting a small business's operations and financial reporting
- Implemented **RBAC and JWT authentication** with Django REST Framework for staff, admin, and guest roles
- Reduced API response latency by **30%** (120 ms) through targeted PostgreSQL indexing in benchmarks

Awards & Publications

- **1st Place, Llama Lounge Agentic AI Hackathon** — Troubadour AI, a context-aware real-time music agent
- **Co-author, IEEE CAI 2025** — "Large Language Models are Autonomous Cyber Defenders" [arXiv:2505.04843](https://arxiv.org/abs/2505.04843)

Education

University of California, Santa Cruz – M.S. Computer Science and Engineering Dec 2025

University of California, Santa Cruz – B.S. Computer Science and Engineering; B.A. Psychology Dec 2022